Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 39

Walks, Paths, and Connectivity

Covers Chartrand & Zhang, Section 12.2

Status of this set

Ungraded practice. Solutions released with it. This is the last practice set of the term.

Part I: How to Practice This Material

The four words, by what may repeat

Term	Vertices repeat?	Edges repeat?
walk	yes	yes
trail	yes	**no**
path	**no**	no
cycle	only start = end	no

A path has no repeated vertex, so it automatically has no repeated edge.

Three mistakes to expect

1. **Calling a trail a path.** Check vertices, not just edges.
2. **Forgetting isolated vertices when counting components.** Each is its own component.
3. **Assuming $n-1$ edges makes a graph connected.** Necessary, not sufficient.

Part II: Practice Problems

Section A — Vocabulary

1. In the graph with $V = \{v_1, \ldots, v_5\}$ and $E = \{v_1v_2, v_2v_3, v_3v_4, v_4v_1, v_2v_5\}$, classify each sequence as a walk only, a trail, a path, or a cycle.

(a) (v_1, v_2, v_3, v_4)

(b) $(v_1, v_2, v_3, v_4, v_1)$

(c) $(v_5, v_2, v_1, v_4, v_3, v_2)$

(d) (v_1, v_2, v_1, v_4)

Section B — Components

2. For each graph, give $k(G)$ and say whether it is connected.

(a) $V = \{1, \ldots, 6\}$, $E = \{12, 23, 31, 45\}$

(b) $V = \{1, \ldots, 5\}$, $E = \{12, 23, 34, 45\}$

(c) $V = \{1, \ldots, 4\}$, $E = \{12, 34\}$

3. A graph has order 9 and size 5.

(a) Can it be connected? Justify.

(b) What is the smallest number of components it could have?

Section C — A proof

4. Prove that if a graph G has order n and every vertex has degree at least $\frac{n-1}{2}$, then G is connected.

Section D — Writing

5. In four to six complete sentences, explain why Theorem 12.26 — that a u–v walk guarantees a u–v path — is useful in practice, and describe the proof technique it uses.